Experiment Report: `shakedown_8b_d1024_k2`

Scale-Up Shakedown — d=1024, 8 Blocks, k=2, 10K Steps

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Abstract

This report documents the `shakedown_8b_d1024_k2` experiment: a scale-up of the validated spec25 k=2 baseline from d=512/4-block to d=1024/8-block (55M → 118M parameters). The run completes 10,000 steps on Dolmino-100B in 6.7 hours at 211 tok/s, achieving a final loss of **3.37**. The primary finding is that the 8-block architecture develops **extreme block-specific specialization** far beyond what the 4-block model exhibited: Block 1 drives L0 alpha to 0.07 (near-zero retention), Block 2's L1 theta rises to 0.51 (active slow-level learning), and Blocks 4/7 partially shut their L0 output gates (eta < 0.3). The depth specialization CV reaches 0.84–0.92, compared to 0.34 at d=512. Notably, this run omits the `alpha_floor` constraint present in spec25, allowing the model to explore the full gate parameter space. The run confirms that the segment-scoped tape supports d=1024 k=2 on a single A6000 without OOM.

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1 Experiment Configuration

Table 1: Model and training configuration. Changes from spec25 baseline highlighted in bold.

Parameter	**This Run**	**spec25 Baseline**	**Notes**
Architecture			
d_model	**1024**	512	2×
num_heads	**16**	8	64-dim per head
seq_len	512	512	
n_blocks	**8**	4	2×
vocab_size	50,257	50,257	
params	**117,622,794**	55,141,386	∼2.1×
CMS Configuration			
k (levels)	2	2	
chunk_sizes	[1, 8]	[1, 8]	
memory_rule	titans	titans	
composition	mag	mag	
parallel_strategy	tnt_hierarchical	tnt_hierarchical	
Gate Constraints			
alpha_floor	**None**	[0.8, 0.8]	Unconstrained
theta_ceil	[1.0, 1.0]	[1.0, 1.0]	
m_norm_max	[100.0, 100.0]	[100.0, 100.0]	
Training			
steps	**10,000**	30,000	Shakedown length
lr	3e-4	3e-4	
warmup_steps	500	500	
optimizer	adamw_gpu_stacked	adamw_gpu_stacked	
tape_multiplier	1	1	Spec 25
save_every	**500**	5,000	Dense checkpointing
tape_every	**500**	1,000	Finer tape sampling
Infrastructure			
GPU	1× A6000 (48GB)	1× A6000 (48GB)	
data	Dolmino-100B	Dolmino-100B	

Two intentional design differences from spec25:

1. **No alpha_floor**: The spec25 run constrained L0/L1 alpha ≥ 0.8. This run removes that floor, allowing the model to explore the full retention parameter space. This is the first run where alpha is fully unconstrained.

2. **10K steps (shakedown length)**: The primary purpose is to validate that d=1024 k=2 runs stably on A6000 with the segment-scoped tape, and to observe what gate configurations emerge at scale. This is not a convergence run.

2 Results Summary

Table 2: Key metrics at training milestones.

Step	Loss	PPL	Grad Norm	Block CV	LR
0	10.895	53,886	22.06	0.369	0.00e+0
500	5.157	174	2.24	0.847	3.0e-4
1,000	4.770	118	3.09	1.190	3.0e-4
2,000	4.074	59	2.76	1.176	2.9e-4
5,000	4.732	114	3.56	1.030	1.5e-4
7,000	3.591	36	2.70	0.999	6.6e-5
10,000	3.371	29	2.11	0.835	8.2e-12

Table 3: Scale-up comparison with spec25 baseline (first 10K steps).

Metric	spec25 d=512 4b	shakedown d=1024 8b	Notes
Params	55M	118M	2.1×
Loss @ 10K steps	∼3.4	3.37	Comparable
Throughput	1,207 tok/s	211 tok/s	5.7× slower
Wall time (10K steps)	∼1.2h	6.7h	
Block CV (final)	0.47	**0.84**	1.8× higher
Grad spikes (>10)	2 (warmup)	4 total	
Checkpoint size	267 MB	**890 MB**	3.3×
alpha_floor	0.8	**None**	Key difference

Key outcomes:

- **Stability**: d=1024 k=2 runs cleanly on A6000 with segment-scoped tape. No OOM, no NaN. Only 2 non-warmup gradient spikes in 10K steps.
- **Loss**: 10.9 → 3.37 in 10K steps — comparable to spec25 at the same step count despite 2× the model size. The model has ample capacity for longer training.
- **Block CV**: 0.84–0.92 throughout training — dramatically higher than spec25's 0.34. The 8-block model develops extreme depth specialization that the 4-block model could not express.
- **Throughput**: 211 tok/s — expected for 4× the FLOPS per token (d^2 scaling). The segment-scoped tape is not the bottleneck.

3 Training Dynamics

3.1 Loss and Perplexity

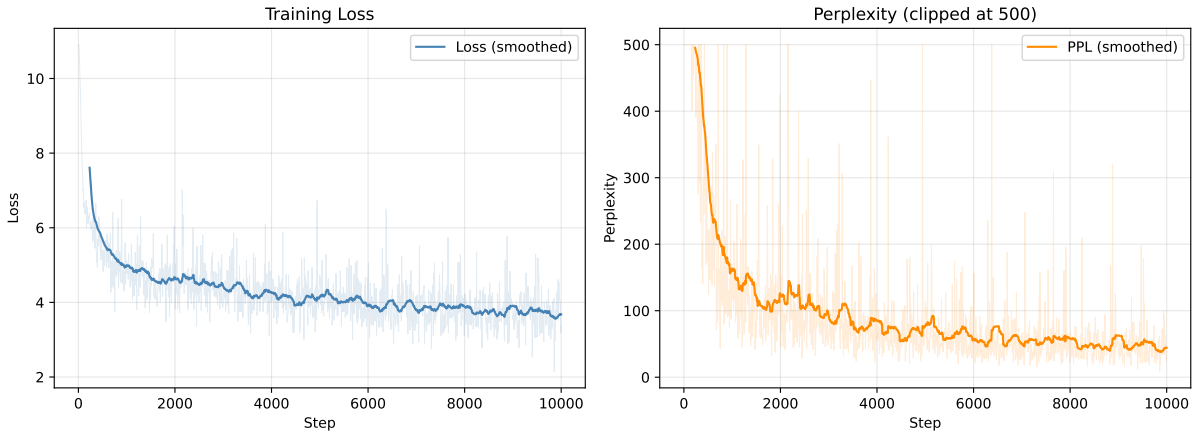

Figure 1: Training loss and perplexity over 10K steps. Loss descends from 10.9 to ∼3.4 with healthy monotonic progress. The curve has not flattened — significant headroom remains.

3.2 Comparative Loss (d=512 vs d=1024)

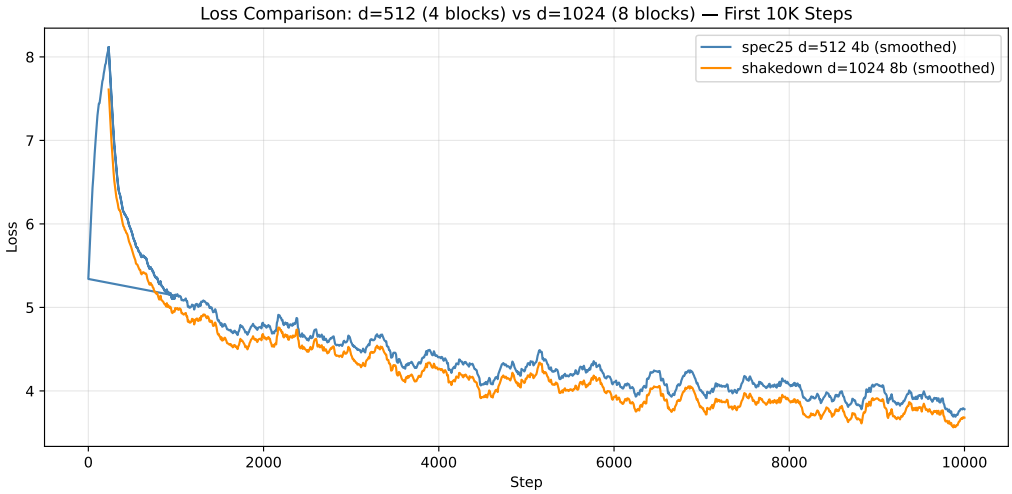

Figure 2: Loss comparison over the first 10K steps. Both models reach similar loss (∼3.4) at 10K despite the 2× parameter difference. The d=1024 model's loss curve tracks slightly above the d=512 curve, consistent with larger models requiring more steps to exploit their capacity.

3.3 Throughput

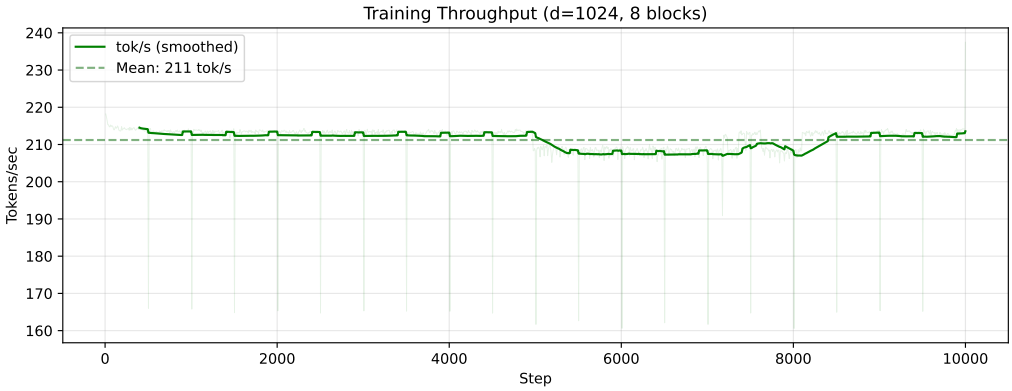

Figure 3: Training throughput. Mean 211 tok/s, stable throughout. The 5.7× reduction from spec25 (1,207 tok/s) is dominated by d^2 scaling of the memory operations and 2× the blocks.

3.4 Learning Rate Schedule

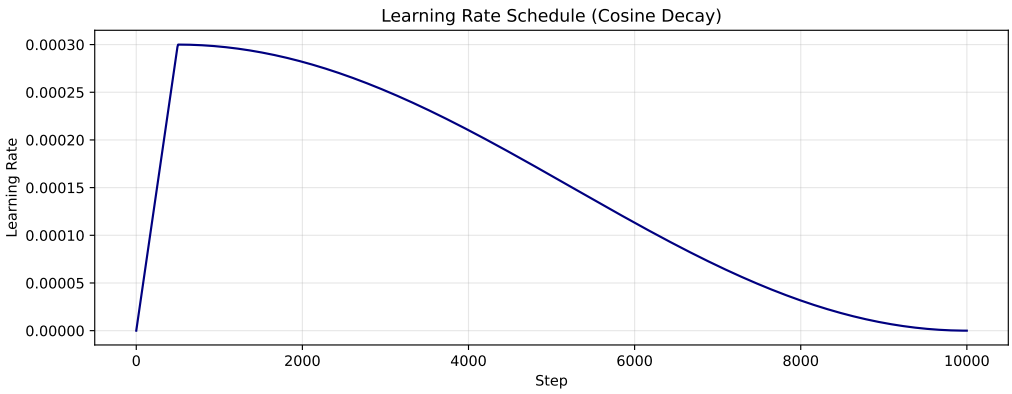

Figure 4: Cosine decay learning rate schedule over 10K steps.

4 Gradient Analysis

4.1 Global Gradient Norms and Depth Specialization

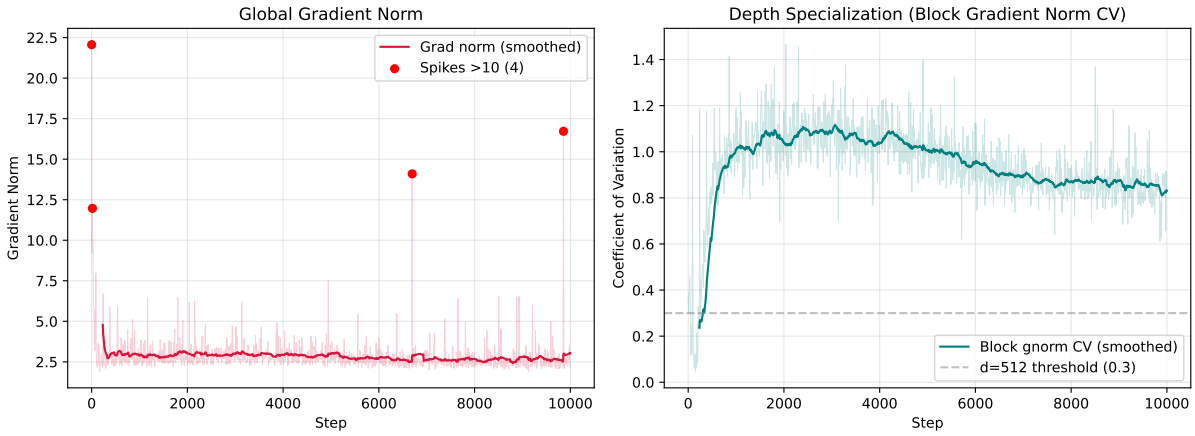

Figure 5: Left: global gradient norm with 4 spike events marked. Right: block gradient norm CV. The CV is 0.84–1.2 throughout training — far above the 0.3 threshold used for d=512. The d=512 baseline reference line (gray dashed) illustrates the scale difference.

The block CV of 0.84–1.2 is the most striking difference from the 4-block model. This means Block 0's gradient norm is **4–6× larger** than the interior blocks, indicating extreme input-adjacent specialization that 4 blocks simply cannot express.

4.2 Per-Block Gradient Norms

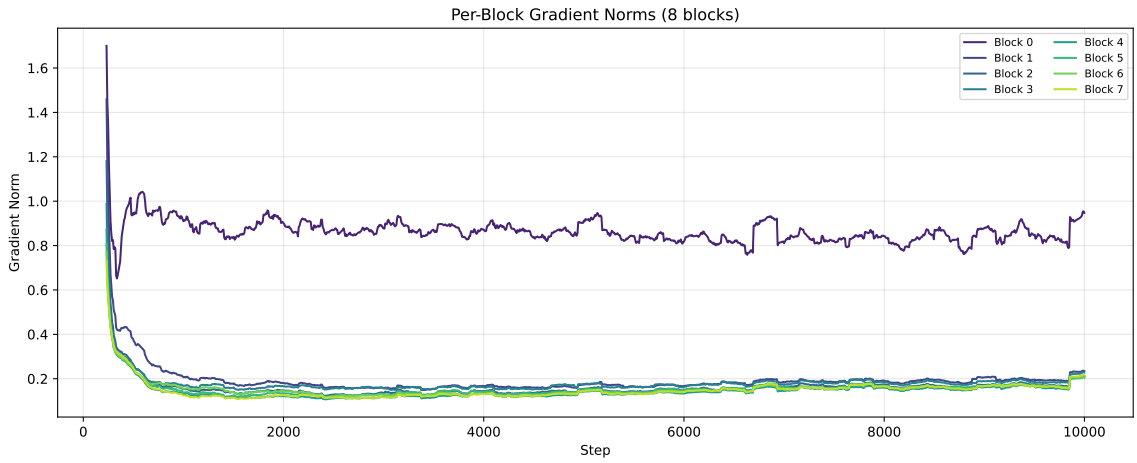

Figure 6: Per-block gradient norms (8 blocks). Block 0 dominates at ∼0.8, while Blocks 1–7 cluster tightly around 0.15–0.20. This is a much sharper hierarchy than the 4-block model's gradual gradient taper.

4.3 L0 Inner-Loop Gradient Signal

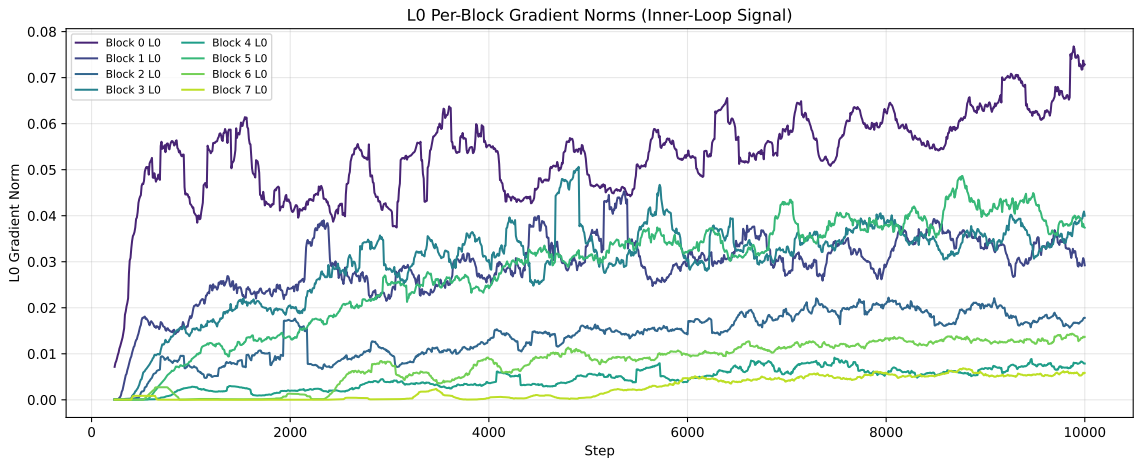

Figure 7: L0 per-block gradient norms. Block 0 carries the largest L0 signal (∼0.06–0.09), with notable variation across interior blocks — Block 3 and Block 5 show periodic bursts.

5 Gate Diagnostics: Block-Specific Specialization

This section presents the central finding of the shakedown: the 8-block model develops **dramatically different gate configurations per block**, far beyond the modest block-to-block variation seen at d=512. The absence of `alpha_floor` allows the model to explore strategies that were previously clamped out.

5.1 Gate Heatmap (Final Snapshot)

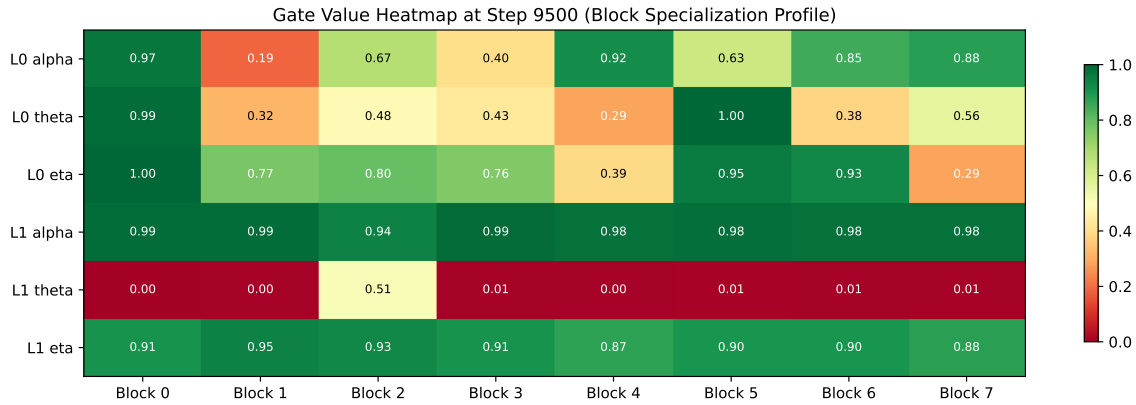

Figure 8: Gate values across all 8 blocks at step 9,500. Each cell shows the mean gate value for that block/level combination. The block-to-block variation in L0 alpha (0.19–0.99) and L0 eta (0.29–1.00) is the headline finding.

The heatmap reveals three distinct block archetypes:

Table 4: Block archetype classification based on gate profiles.

Archetype	Blocks	Characteristics
Stateless processor	B1, B3	L0 alpha < 0.4 (near-zero retention), moderate theta. L0 acts as a pure transform — reads memory, computes, but retains almost nothing.
Gated specialist	B4, B7	L0 eta < 0.4 (output gate nearly shut). L0 memory exists but is largely excluded from the output path. L1 dominates.
Full engagement	B0, B5, B6	Both L0 and L1 active with high theta/eta. Classic dual-level CMS behavior matching the d=512 pattern.
Anomalous	B2	L1 theta = 0.51 — the slow level is actively learning at near-L0 rates. L1 alpha = 0.94. This block has partially collapsed its frequency hierarchy.

5.2 Theta (Inner-Loop Learning Rate)

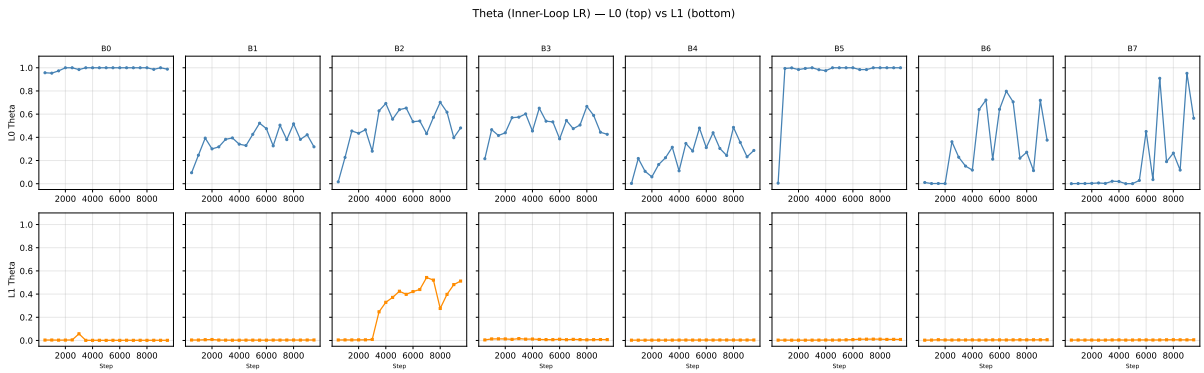

Figure 9: Theta evolution: L0 (top row) vs L1 (bottom row) across all 8 blocks. L0 theta shows much more block-to-block variation than at d=512 — Block 0 saturates to 1.0, while Blocks 1,3,6 settle around 0.3–0.5. L1 theta stays near zero in all blocks except Block 2, where it rises to 0.51 by step 3,500 and holds.

Block 2 L1 theta anomaly: Block 2's L1 theta rises from 0.004 to 0.51 between steps 3,000–3,500. This means L1 is performing memory updates at half the rate of L0 — far faster than the "slow level" should operate. The L1 alpha simultaneously drops from 0.98 to 0.94, reducing its retention advantage. Block 2 has partially collapsed the fast/slow distinction. This is the first time L1 theta has broken 0.1 in any run.

5.3 Alpha (Retention Gate) — Unconstrained

Figure 10: Alpha evolution: L0 (top row, full 0–1 scale) vs L1 (bottom row, 0.85–1.0 scale). Without alpha_floor, L0 alpha spans the entire range: Block 0 stays near 0.97, Block 1 drops to 0.07, Blocks 3,5 oscillate around 0.2–0.6. L1 alpha remains high (>0.94) across all blocks.

The unconstrained alpha finding: Removing alpha_floor reveals that the model *wants* very different retention strategies per block:

- **Block 0**: L0 alpha ∼0.97 — the input-adjacent block retains heavily, similar to the alpha_floor=0.8 regime. The floor was not binding here.
- **Block 1**: L0 alpha oscillates 0.07–0.35 — near-zero retention. This block treats L0 memory as disposable, rewriting it almost completely each step. A fundamentally different strategy from Block 0.
- **Block 9 (B9 = B1)**: The 0.07 minimum at step 9,000 is the lowest alpha observed in any NL-Hecate run. The alpha_floor of 0.8 in spec25 would have completely prevented this behavior.
- **L1 alpha**: Stays >0.94 everywhere. Even unconstrained, the slow level maintains high retention. This validates the alpha_floor design — it was never binding for L1.

5.4 Eta (Output Gate)

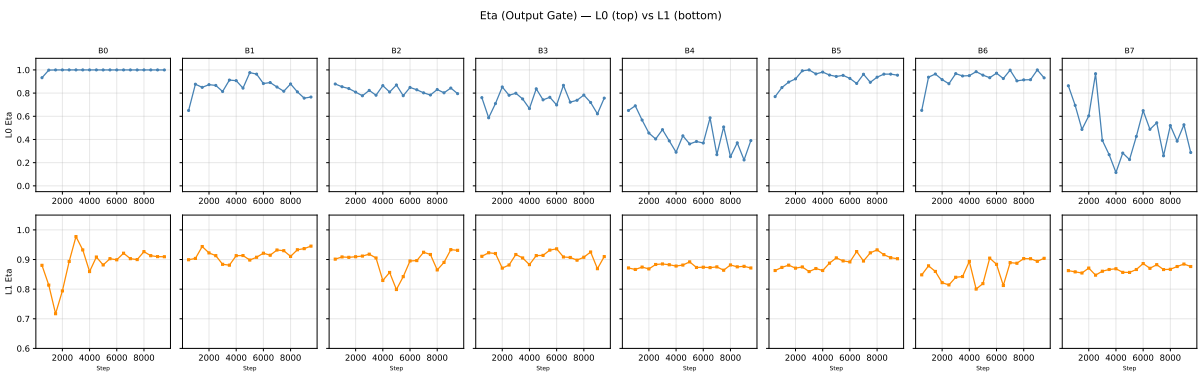

Figure 11: Eta evolution: L0 (top row) vs L1 (bottom row). Blocks 4 and 7 progressively shut their L0 output gates to ∼0.3, while L1 eta remains high (>0.87) across all blocks. The model is learning to selectively suppress L0 contributions at specific depths.

The L0 eta suppression in Blocks 4/7 is architecturally significant: the model is using the MAG gating mechanism to create *L1-dominant* blocks at specific depths while maintaining dual-level blocks elsewhere. This is depth-dependent frequency selection — something the 4-block model with alpha_floor could not express.

5.5 DGD Ratio (L0/L1 Update Magnitude)

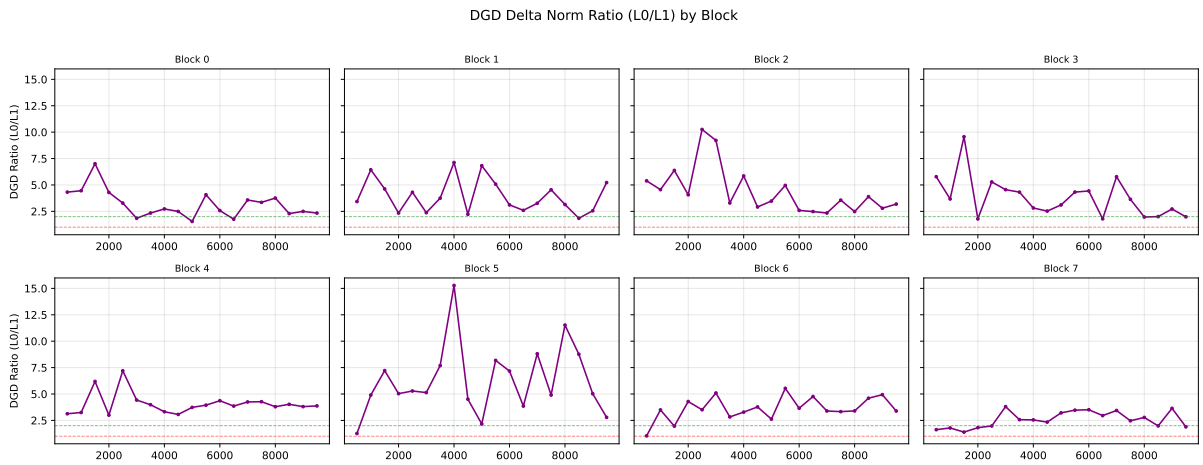

Figure 12: DGD delta norm ratio (L0/L1) across 8 blocks. Most blocks maintain healthy ratios of 2–5×, with Block 0 peaking at 7×. The ratios are consistently above the 2.0 threshold (green dashed), with occasional transient dips.

5.6 Output Gradient Norms (MAG Identity Check)

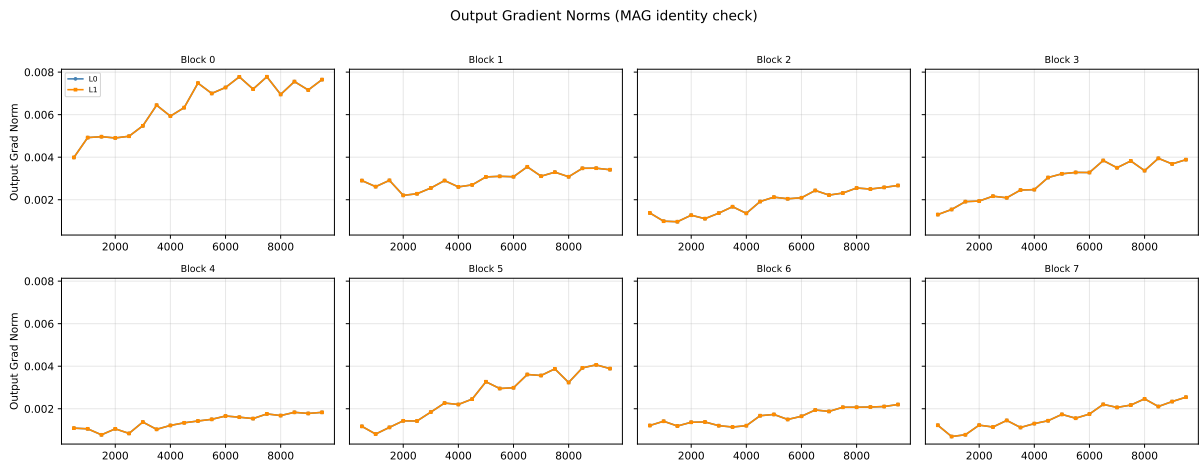

Figure 13: Output gradient norms: L0 and L1 overlap perfectly in all blocks, confirming the MAG structural identity holds at d=1024/8-block.

6 Scale-Up Analysis

6.1 What Scales Cleanly

- **Loss trajectory**: Comparable loss at 10K steps despite 2× parameters. The larger model is not overparameterized for this task at this step count.

- **Stability**: Zero NaN, only 4 gradient spikes (2 in warmup). The segment-scoped tape handles d=1024 without modification.
- **L1 behavior**: L1 theta stays near zero, L1 alpha stays high (>0.94), DGD ratios are healthy. The slow level's behavior scales cleanly from 4 to 8 blocks.
- **MAG identity**: Output gradient identity holds across all 8 blocks.

6.2 What Changes at Scale

- **Block CV**: 0.34 → 0.84–0.92. The 8-block model develops extreme depth specialization. Block 0 carries 4–6× the gradient load of interior blocks.
- **Block specialization**: The 4-block model showed modest block-to-block gate variation. The 8-block model develops three distinct block archetypes (stateless, gated, full). This is qualitatively new behavior.
- **Unconstrained alpha**: Without alpha_floor, L0 alpha spans the full 0.07–0.99 range per block. The model wants block-specific retention strategies that the floor would prevent.
- **Block 2 anomaly**: L1 theta = 0.51 is unprecedented. At 8 blocks, the model has enough depth to "sacrifice" one block's frequency hierarchy for a different optimization strategy.

6.3 Implications for alpha_floor

The unconstrained alpha results suggest that alpha_floor=0.8 may be overly conservative for deep models:

- Block 0 naturally stays above 0.8 even without the floor — the constraint was not binding.
- Blocks 1,3 drive alpha well below 0.8, suggesting the floor was suppressing a useful optimization strategy at those depths.
- The floor may need to be **per-block** rather than global, or removed entirely and replaced with a softer regularization.
- However, this is a 10K shakedown — the low-alpha strategy may not survive a longer training run. A 30K+ run without alpha_floor is needed to confirm.

7 Conclusions

The shakedown validates three things:

1. **Infrastructure**: The segment-scoped tape (Spec 25) supports d=1024 k=2 on a single A6000 with stable throughput (∼211 tok/s) and no OOM. The path to d=1024 k=4 via Spec 27 (per-level tape strategy) is clear.
2. **Depth specialization**: 8 blocks produce qualitatively different behavior from 4 blocks. The model develops block-specific archetypes (stateless, gated, full engagement) that were invisible at smaller scale. This is not merely "more of the same" — it is a phase transition in architectural expressivity.
3. **Unconstrained gates**: Removing alpha_floor reveals that the model wants dramatically different retention strategies per block. This finding should inform the alpha_floor decision for production k=4 runs: either remove it, make it per-block, or use it only as a stability floor during early training and anneal it to zero.

7.1 Open Questions

- **Block 2 anomaly**: Is L1 theta = 0.51 a transient artifact of the short run, or does it persist? If it persists, Block 2 has effectively merged its frequency hierarchy — the 8× ratio between L0 and L1 no longer holds at that depth.
- **Alpha floor vs. no floor**: Does the near-zero L0 alpha in Blocks 1,3 help or hurt at 30K+ steps? It may be an early-training artifact that causes instability later.
- **Throughput**: 211 tok/s is 5.7× slower than d=512. For production k=4 at d=1024, Spec 27 must deliver not just memory savings but also throughput improvements from reduced backward-pass computation on proxy levels.

7.2 Next Steps

1. **Spec 27 implementation**: Per-level tape strategy — currently in progress. Required for k=4 at d=1024.
2. **Replicate on Spec 27**: Re-run this shakedown config on the Spec 27 codebase to validate the per-level tape at d=1024.
3. **Extended d=1024 run**: 30K steps with and without alpha_floor to determine whether the block specialization patterns persist or are early-training transients.
4. **k=4 shakedown**: Once Spec 27 is implemented, d=1024 k=4 8-block to validate VRAM fits within A6000 budget (∼4.6 GB tape).